SYLLABUS
STATISTICAL LEARNING

UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

Course Title: Statistical Learning

Instructor: Sabyasachi Chatterjee

Institution: University of Illinois Urbana-Champaign

Lecture Hours: TBA

Office Hours: TBA

Course Dates: TBA

Course Website: TBA

Lecture notes, announcements, assignments, and supplementary materials will be posted on the course website.

Course Description and Goals:

This is a graduate course in statistical learning intended primarily for Master's students and first- or second-year Ph.D. students. The course will develop statistical learning largely through the perspective of Bayesian statistics and probabilistic modeling.

Statistical learning is often presented as a collection of algorithms for prediction and data analysis. This course takes a different viewpoint. We will study statistical learning through the lens of probabilistic inference. We will view statistical and machine learning methods as arising from probabilistic models, typically parametrized by a function class, and then study how inference can be carried out given data.

Bayesian statistics will serve as a guiding principle throughout much of the course. We will study how uncertainty is represented, updated, and propagated through probabilistic models. At the same time, we will discuss important frequentist tools such as cross-validation, bootstrap, and prediction error estimation.

The goal is not to memorize a laundry list of algorithms. Rather, the goal is to understand a coherent set of ideas connecting statistics, probability, computation, and modern machine learning.

The course is organized around a few major themes:

- Statistical learning and function estimation.
- Bayesian modeling and inference.
- Bayesian computation and uncertainty quantification.

- Modern generative AI.

Students will come away seeing modern statistical learning and AI not as disconnected collections of algorithms, but as developments of ideas from statistics, probability, Bayesian inference, computation, and function approximation.

Prerequisites:

Students should be comfortable with probability, mathematical statistics, linear algebra, and multivariable calculus at the undergraduate level. Prior exposure to regression and statistical modeling will be helpful. Basic programming experience in Python is helpful but not required. The course will include computational demonstrations and examples using Python notebooks. Students are free to use LLMs for coding exercises in the course.

Learning Objectives:

By the end of the course, students should be able to:

- understand statistical learning from a probabilistic perspective;
- formulate regression and classification problems using statistical models;
- formulate and analyze Bayesian models;
- understand Bayesian inference in simple and moderately complex models;
- understand empirical Bayes methods and shrinkage;
- understand the role of regularization in high-dimensional regression;
- apply resampling methods such as cross-validation and the bootstrap;
- understand nonlinear regression and function estimation methods;
- understand the foundations of Bayesian computation;
- implement and analyze basic MCMC and variational inference algorithms;
- understand the basic principles behind neural networks;
- understand the key ideas behind modern generative AI models such as variational autoencoders, diffusion models, and transformers.

Topics:

The following list is tentative and may be adjusted depending on the pace of the course.

I. Bayesian Foundations and Supervised Learning

- Bayesian philosophy and probabilistic modeling
- Prior, likelihood, posterior, and posterior predictive distributions
- Linear regression
- Logistic regression
- Generalized linear models
- Frequentist and Bayesian perspectives on regression
- Classification using LDA and QDA

II. Model Assessment and Resampling

- Cross-validation
- Bootstrap methods

III. Bayesian Inference and Empirical Bayes

- Conjugate Bayesian models
- Gaussian sequence models
- Empirical Bayes methods
- James–Stein estimation and shrinkage
- Dirichlet–multinomial models
- Bayesian bootstrap

IV. Regularization and High-Dimensional Models

- Ridge regression
- Bayesian interpretation of ridge regression
- Lasso, Coordinate Descent.

V. Nonlinear Function Estimation

- Smoothing splines
- Local polynomial regression
- Gaussian process regression
- Trend filtering
- Isotonic regression

VI. Bayesian Computation

- Monte Carlo methods
- Metropolis–Hastings
- Gibbs sampling
- Hamiltonian Monte Carlo
- Variational inference
- ELBO optimization
- Connections between EM and variational methods

VII. Modern Generative AI

- Neural network fundamentals
- Latent variable models
- Variational autoencoders
- Diffusion denoising probabilistic models
- Transformers

Course Materials:

Lecture notes will be provided for every class. The course will also include Python notebooks and computational demonstrations. We will fit models, run algorithms, visualize outputs, and examine examples in class.

References:

There is no required textbook for the course. Lecture notes will be the primary source of material. The following references could be useful.

(1) Aditya Guntuboyina's lecture notes and course materials for *STAT 238: Bayesian Statistics* at UC Berkeley: https://stat238.berkeley.edu/spring-2026/.
(2) James, Gareth, Daniela Witten, Trevor Hastie and Robert Tibshirani, *An Introduction to Statistical Learning: With Applications in Python* (ISLP), Springer.
(3) Bishop, Christopher M. and Hugh Bishop, *Deep Learning: Foundations and Concepts*, Springer.

Evaluation:

The tentative grading scheme is:

- Homework Assignments: 50%
- Final Project: 50%

The final project may be completed in groups of ≤ 3 members. Some project ideas are given at the end of this document.

Homework:

Homework assignments are intended to help students engage with both the mathematical and computational aspects of the course. Assignments may include written exercises, conceptual questions, and programming problems.

Collaboration is allowed, but each student must write up their own solutions and submit their own code. Students should clearly indicate the names of any classmates with whom they collaborated.